MILITARY SPECIFICATION

TEST SET, GUIDED MISSILE AN/DSM-68

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The test equipment covered by this specification shall be for shop use and shall be used to perform preflight tests on the AGM-45A-1, -1A, -2, -3, -4, and -5 (SHRIKE) missile.

1.2 Classification - The equipment covered by this specification shall consist of the following items:

Type Designation	Applicable Paragraph
Test Set, Guided Missile TS-1976A/DSM-68	3.5.1
Case, Test Set CY-4082/DSM-68	3.5.2
Cover, Test Set CW-712/DSM-68	3.5.2
Simplified Test Instruction Card (63A3E32)	3.5.3
Hose Assembly, Pneumatic Output (63A3D58)	3.5.4
Adapter Assembly, Pneumatic Hose (63A3B50)	3.5.5
Cable Assembly, W1 of AN/DSM-68 (63A3D70)	3.5.6
Cable Assembly, Power, W2 of AN/DSM-68 (64A53D2)	3.5.7
Hose, Pneumatic Input, P/O AN/DSM-68 (64A53D1)	3.5.8
Adapter Assembly, Pneumatic Supply to Input Hose (64A53B14)	3.5.9
Simulator, Radar Signal SM-416A/DSM-68	3.5.10
Case, Shipping and Storage, Test Set CY-4647/DSM-68	3.5.11
Wrench, Hexagon Key (58A5A294-2)	3.5.12

FSC-4935

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2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of the invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-T-5422	Testing Environmental. Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for
MIL-W-27076	Workmanship Standards for Ground Electronics and Associated Equipment

STANDARDS

Military

MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-470	Maintainability Program Requirements for Systems and Equipments Prediction
MIL-STD-471	Maintainability Demonstrations
MIL-HDBK-472	Military Standardization Handbook, Maintainability Predictions
MIL-STD-781	Reliability Tests, Exponential Distribution

DRAWINGS

Naval Air Systems Command

DL 63A3F1	Data List for AN/DSM-68, and all drawings therein

PUBLICATIONS

Military

ANA Bulletin No. 400	Electronic Equipment; Aircraft and Guided Missiles, Applicable Documents

Naval Air Systems Command

WS-5491	Description and Requirements; Simulator, Radar Signal SM-418A/DSM-68

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2.1.1 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications and documents required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

(2) Copies of DL 63A3F1, and the drawings therein may be obtained upon application to the Commanding Officer, Naval Air Technical Services Facility, 700 Robbins Avenue, Philadelphia, Pennsylvania 19111.

(3) Copies of Purchase Description WS-5491 may be obtained upon application to the Commander, Naval Weapons Center, China Lake, California 93555.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials -

3.2.1 General Requirements - The parts and materials specified in the drawings and related documents for use in construction of this equipment were selected to ensure the fulfillment of all requirements of this specification. Any deviation by the contractor from the use of these specified parts and materials necessarily places both the burden of proof for the acceptability of the change, and the cost associated therewith, upon the contractor. Approval for deviations must be requested from the procuring activity by the contractor and granted by the procuring activity prior to submission of equipments for preproduction tests. Approval of parts and material deviations shall not be construed by the contractor to constitute a waiver of the performance requirements or any other requirements of this specification.

3.2.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained by the contractor as outlined in MIL-T-21200.

3.2.2 Supplemental Requirements - In the event that the contractor proposes to alter the equipment design delineated by this specification and applicable drawings, the following supplemental requirements for the selection of parts and materials are applicable.

3.2.2.1 Basic Conformance - Parts and materials selected for use in the equipment shall conform to the requirements of MIL-T-21200. Parts shall be selected from those parts included in ANA Bulletin No. 400, unless the parts listed therein will not perform adequately in the intended applications and environments specified herein.

3.2.2.2 Interchangeability - The equipment shall meet the interchangeability requirements of MIL-T-21200. The drawings listed in DL 63A3F1 shall be the standard for measuring interchangeability of all applicable units, assemblies, subassemblies, and parts.

3.3 Design and Construction - The equipment shall be manufactured in accordance with the drawings and subsidiary specifications listed in DL 63A3F1 and related lists, except insofar as errors of omission or commission on the drawings may be incompatible with the requirements of this and related specifications. In any case it is a condition of final acceptance that the equipment manufactured meet all the performance requirements of this specification, and such acceptance test documents as may be applicable. Proposed changes, either to correct drawing discrepancies or to effect improvements, shall be specifically approved by the procuring activity prior to their incorporation.

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3.3.1 Reliability -

3.3.1.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 150 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.1.2 Operating Life - The equipment shall have a minimum total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment and, the wearout life of such parts, shall be determined by the contractor and submitted to the procuring agency.

3.3.1.3 Reliability in Mean-Time-Between-Failures (MTBF) - The equipment shall have a minimum of 600 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.2 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be within the limits of MIL-I-6181 except that broadband short duration switching transient interference shall not require limits.

3.3.3 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (30 ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient
Input power voltage	115 ± 1.5 volts, 400 ± 5 Hz

3.3.4 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for class 3 equipment, except as modified herein.

3.3.4.1 Shock - In addition to the shock requirements of MIL-T-21200 the equipment, when stored in its shipping and storage case, shall operate as specified in 3.5 after subjection to two half-sine-wave, 70g shocks, each of 7 milliseconds duration, applied upward along the vertical axis of the equipment case. Physical damage to the equipment is permissible, provided the operational requirements of 3.5 are maintained.

3.3.5 Warm-Up Time - The equipment, excluding the radar signal simulator SM-416A/DSM-68, shall require no warm-up time under the service conditions specified in 3.3.4.

3.3.6 Primary Input Power Requirements - The equipment shall be capable of satisfactory operation when supplied input power having characteristics as defined in MIL-T-21200. The power required shall be of the following type and shall not exceed the specified amount.

AC Power, Three-Phase, Four-Wire Wye, 115-V (line to ground) 380-420 Hz, 200 VA

3.3.7 Mean-Time-To-Repair (MTTR) - The mean-time-to-repair (MTTR) of the equipment shall not be greater than 1.5 hours, when replacing assemblies and parts that are intended for replacement at the organizational level. Only where design changes shall cause significant deviations from procuring activity drawings, a prediction of MTTR shall be made in accordance with MIL-STD-470 and MIL-HDBK-472 procedure II, and shall be demonstrated in accordance with 4.2.2.

3.4 Performance - The performance of the equipment shall be as set forth below plus any and all performance requirements stated on the drawings of DL 63A3F1. Unless otherwise specified, these requirements shall apply under both standard and extreme service conditions. When reduced performance requirements for service conditions are acceptable, such variations are specified.

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3.5 Detail Requirements -

3.5.1 Test Set, Guided Missile TS-1976A/DSM-68 - The test set shall be manufactured in accordance with the documents listed on DL 63A3F1, and shall meet the requirements stated thereon and in the following subparagraphs.

3.5.1.1 Function - The test set shall provide electrical and pneumatic inputs to the missile being tested, and shall monitor the electrical outputs of the missile being tested, as detailed herein.

3.5.1.2 Electrical Requirements - The test set shall meet the following electrical requirements:

3.5.1.2.1 TEST METER Indications - Refer to table 1-1. For each position of the SELECTOR switch, and each corresponding input to J2 pins, the TEST METER indication shall be on the center of the line which determines the area of the GOOD zone as called out in the table.

Table 1-1. TEST METER Indications

SELECTOR Switch Position	Input to J2 Pins	Input	TEST METER Indication
2	12 (+) 18 (-)	0.500 ± 0.003 ma d-c	Good (center) **
CAL *			Good (center) ***
1 *	14 18	48 ± 2.8 ohms	Good (right)
1 *	14 18	64 ± 2.8 ohms	Good (left)
2	12 (+) 18 (-)	57.5 ± 5.7 mv d-c	Good (left)
2	12 (+) 18 (-)	77.7 ± 6.7 mv d-c	Good (right)
3	13 (+) 18 (-)	57.5 ± 5.7 mv d-c	Good (left)
3	13 (+) 18 (-)	77.7 ± 6.7 mv d-c	Good (right)

* Requires 115 volt 400 Hz 3ø power at J1.
** Adjust the meter zero adjust screw to obtain this indication
*** Adjust R4 to obtain this indication.

3.5.1.2.2 TARGET POSITION Meter Indications - Refer to table 1-2. For each input to J2 pins, the TARGET POSITION meter indication shall be on the center of the line which determines the readings as called out in the following table.

3.5.1.2.3 Radar Simulator Power - With electrical power (115 ± 11.5 volts a-c, 400 ± 20 Hz, 3-phase) applied to connector J1, the output at pin 3 of RADAR SIMULATOR connector J3 shall be 28 ± 4 volts d-c, with a maximum ripple of 6 volts peak-to-peak. For these measurements the radar signal simulator shall be connected to terminal J3, and the PNEU SUPPLY switch shall be alternately ON and OFF.

3.5.1.2.4 Loudspeaker Indicatications - With electrical power (2.5 ± 0.01 volts a-c, 2000 Hz, 1-phase) applied to connector J2, between pins 16 and 18, the loudspeaker indications shall be an audible tone.

3.5.1.2.5 Missile Power Output - (a) With electrical power (115 volts a-c, 400 Hz,

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3-phase) applied to connector J1, and with J2 pin 4 connected to J2 pin 18 (ground), and with MSL PWR switch ON, the output at connector J2, pin 1 shall be within ±0.3 volt a-c of the input voltage at connector J1, pin 1. (b) With electrical power applied as in (a) and with the connector (pin 4 to pin 18) removed, or the MSL PWR switch OFF, there shall be no output present.

Table 1-2. TARGET POSITION Meter Indications

Input To J2 Pins		Input	Meter Indication
+	-		
8	9	11.5 ± 2.8 v d-c	Climb 90
9	8	11.5 ± 2.8 v d-c	Dive 90
10	11	10.3 ± 2.5 v d-c	Roll Left 9
11	10	10.3 ± 2.5 v d-c	Roll Right 9

3.5.1.2.6 ELAPSED TIME Meter Indication - The ELAPSED TIME meter shall indicate the total Power-On time for the test set.

3.5.1.2.7 PWR IND Lamp Indication - With electrical power (115 volts a-c, 400 Hz, 3-phase) applied to connector J1, and with PWR switch ON, the PWR IND lamp shall be ON. Removal of electrical power, or returning PWR switch to OFF, shall cause the PWR IND lamp to be OFF.

3.5.1.3 Pneumatic Requirements - With electrical power (115 volts a-c, 400 Hz, 3-phase) applied to connector J1, and with PWR and PNEU SUPPLY switches ON, the test set shall meet the following pneumatic requirements:

3.5.1.3.1 Pneumatic Pressure Supply - The pneumatic pressure supply shall consist of a 250 ± 25 psig dry air or nitrogen source, with 10 cfm minimum delivery and a dew point of -32°C maximum. The air or nitrogen shall be filtered for non-contamination to a particle size of 50 microns maximum, and shall be oil-free.

3.5.1.3.2 Minimum Flow Rate Test - With the pneumatic pressure supply connected to the test set via pneumatic input and output hoses attached to input and output couplings, and the input pressure adjusted so that the PNEUMATIC INPUT gauge indication is at the left edge of the GOOD zone the PNEUMATIC OUTPUT gauge indication shall be within the GOOD area for the following flow rates:

(1) 2.0 ± 0.2 scfm, or

(2) 1.2 (+0.2, -0.0) scfm

3.5.1.3.3 Maximum Flow Rate Test - With the above connections, and the input pressure increased so that the PNEUMATIC INPUT gauge indication is at the right edge of the GOOD zone, the PNEUMATIC OUTPUT gauge indication shall be within the GOOD zone for a flow rate of 2.0 ± 0.2 scfm.

NOTE: (1) With the PNEU SUPPLY switch turned OFF the pneumatic supply solenoid shall de-energize and the flow rate shall drop to zero.

(2) With PNEU SUPPLY switch turned ON the pneumatic supply solenoid shall energize and the flow rate shall be restored to 2.0 ± 0.2 scfm.

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3.5.1.3.4 Pressure Leak Test - With one end of the pneumatic output hose connected to the output coupling (and the other end disconnected), and the input pressure increased so that the PNEUMATIC INPUT gauge indication is in the middle of the GOOD zone, the PNEUMATIC OUTPUT gauge indication shall be within or above the GOOD area.

NOTE: With the pneumatic pressure supply closed for 5 minutes to prevent additional input the PNEUMATIC OUTPUT gauge indication shall remain within, or above the GOOD zone.

3.5.2 Case, Test Set CY-4082/DSM-68 and Cover, Test Set CW-712/DSM-68 - The case and cover shall be manufactured in accordance with the documents listed on DL 63A3F1 and shall meet the requirements stated thereon.

3.5.2.1 Function - The case and cover shall be used to house the following items:

Test Set TS-1976A/DSM-68

Umbilical Cable Assembly

Pneumatic Output Hose

Pneumatic Hose Adapter

Hexagon Wrench

Spare Fuses

Simplified Test Instructions Card

Handbook, Operation and Service Instructions

3.5.3 Test Instructions (Simplified) Card (63A3E32) - The simplified test instructions card shall be manufactured in accordance with the documents listed on DL 63A3F1, and shall meet the requirements stated thereon.

3.5.3.1 Function - The card provides ruggedized and abbreviated test instructions for use by operating personnel.

3.5.4 Hose Assembly, Pneumatic Output (63A3D58) - The pneumatic output hose assembly shall be manufactured in accordance with the documents listed on DL 63A3F1, and shall meet the requirements stated thereon.

3.5.4.1 Function - The output hose is used to connect the pneumatic output coupling of the test set to the hose adapter of the missile system being tested.

3.5.5 Adapter Assembly, Pneumatic Hose (63A3B50) - The adapter shall be manufactured in accordance with the documents listed on DL 63A3F1, and shall meet the requirements stated thereon.

3.5.5.1 Function - The adapter is used to connect the output hose to the missile system being tested.

3.5.6 Cable Assembly (W1) (63A3D70) - Cable (W1) shall be manufactured in accordance with the documents listed on DL 63A3F1, and shall meet the requirements stated thereon.

3.5.6.1 Function - Cable (W1) is used to electrically connect the test set and the umbilical connector of the missile system being tested.

3.5.7 Cable Assembly, Power (W2) (64A53D2) - Cable (W2) shall be manufactured in accordance with the documents listed on DL 63A3F1, and shall meet the requirements stated thereon.

3.5.7.1 Function - Cable (W2) is used to electrically connect the electrical power source and the POWER IN connector of the test set.

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3.5.8 Hose, Pneumatic Input (P/O) (64A53D1) - The input hose shall be manufactured in accordance with the documents listed on DL 63A3F1, and shall meet the requirements stated thereon.

3.5.8.1 Function - The input hose is used to connect the pneumatic input coupling of the test set and the pneumatic pressure supply, or the pneumatic supply adapter.

3.5.9 Adapter Assembly, Pneumatic Supply (63A53B14) - The adapter shall be manufactured in accordance with the documents listed on DL 63A3F1, and shall meet the requirements stated thereon.

3.5.9.1 Function - The adapter is used to connect certain pneumatic pressure supplies and the input hose.

3.5.10 Simulator, Radar Signal SM-416A/DSM-68 - The simulator shall be manufactured in accordance with the documents listed on DL 63A3F1, and shall meet the requirements stated thereon.

3.5.10.1 Function - The simulator provides radar signals as required to check out the missile system being tested.

3.5.11 Case, Shipping and Storage, Test Set CY-4647/DSM-68 - The shipping and storage case shall be manufactured in accordance with the documents listed on DL 63A3F1, and shall meet the requirements stated thereon.

3.5.11.1 Function - The shipping and storage case provides a protective container for both shipment and storage of the following items.

Test Set AN/DSM-68

Power Cable

Pneumatic Input Hose

Pneumatic Supply Adapter

Radar Signal Simulator

3.5.12 Wrench, Hexagon Key (58A5A294-2) - The wrench shall be manufactured in accordance with the documents listed on DL 63A3F1, and shall meet the requirements stated thereon.

3.5.12.1 Function - The wrench is used to loosen and tighten the set screws in the knob of the SELECTOR switch as a part of test set maintenance.

3.6 Workmanship - General workmanship shall be in accordance with MIL-W-27076 and MIL-STD-454, requirement 9. Soldering shall be in accordance with MIL-STD-454, requirement 5.

3.7 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or order (see 6.4).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the Specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction Tests

(2) Initial Production Tests

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(3) Acceptance Tests

(4) Life Test

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The Government Inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition (see 6.4).

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the requirements of MIL-T-5422 to the limits specified in MIL-T-21200 for class 3 equipment except as modified by 3.3.4.1. In addition preproduction tests shall include interference tests in accordance with MIL-I-6181, drop, and water-tightness tests in accordance with MIL-T-21200. When required (see 3.3.7) Maintainability Demonstration tests shall be in accordance with MIL-STD-471.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall by by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment (see 6.4).

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

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4.3.4 Reconditioning of Initial Production Test Sample - On completion of the Initial Production Test the equipment shall be reworked by the contractor by replacing all "wear" or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitive results for all acceptance tests (see 6.4). Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests
(3) Reliability Assurance Tests
(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product
(2) Operational Test
(3) Manufacturing Run-In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation (see 6.4).

4.4.1.3 Manufacturing Run-In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration for the purpose of detecting flaws and imperfect workmanship. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equip-

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ment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action.) Equipments shall be selected for sampling tests by the Government Inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	0 (see note)
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

NOTE: If by contract action the Initial Production Test (see 4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first ten produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run-in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipments.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests shall be conducted as required by both the Qualification Phase and Sampling Phase.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in table 5 of MIL-STD-781. For the Qualification Phase Test Level E shall be used except lower temperature shall be 0° Centigrade. The Accept-Reject Criteria for Test Plan VI shall be used.

4.4.3.2 Reliability Sampling Phase Tests - Samples of the equipment shall be tested as outlined in MIL-STD-781, under the section entitled "Sampling Phase of Production Reliability Tests". For the Sampling Phase Test Level E shall be used, except lower temperature shall be 0° Centigrade. The Accept-Reject criteria for Test Plan VI shall be used to determine the length of the tests (until an accept or reject decision is reached).

4.4.3.2.1 Lot Size for Sampling Phase - The entire contract quantity (with the exception of those equipments used in the Qualification Phase) shall be considered one lot. The test program shall start the first month after the Qualification Phase has been completed. Samples shall be selected and placed on test in the same manner as though 1 month's production was a lot. The samples

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shall be tested until an accept or reject decision is reached. In the meantime other equipments constructed shall be shipped. The test results shall be summarized monthly for the procuring activity (see 6.4). The procuring activity reserves the right to stop the acceptance of equipment any time after one or more reject decisions have been reached, pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment quality control, etc., so that the entire contract quantity will show an accept decision.

4.4.3.3 Test Details - Test details such as the length of the Test Cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, and preventive maintenance to be allowed during the test, shall be part of the test procedures to be submitted by the contractor and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment, after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance, or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests (see 6.4).

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual tests. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government Inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Tests, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1

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Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the standard conditions specified in 3.3.3.

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the Government Inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test (see 6.4). In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests, and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval (see 6.4). The right is reserved by the procuring activity or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests when deemed necessary.

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4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all "wear" or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and be resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector (see 6.4).

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 (see 6.4).

6. NOTES

6.1 Intended Use - The intended use of the test set is to check out the operations of the SHRIKE missile components as follows:

Electrical system

Pneumatic pressure system

Radar receiver

Control surfaces (mechanical operation)

6.2 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in DL 63A3F1, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) The drawings of DL 63A3F1 - The drawings shall have precedence over other referenced specifications. Any deviation from the drawings shall be specifically approved by the procuring activity.

(4) Referenced Specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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6.4 Ordering Data - Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(1) Title number and date of this specification
(2) Nonstandard parts and material approval (see 3.2.1.1)
(3) Parts replacement data (see 3.3.1.2)
(4) Design data (see 3.7)
(5) Preproduction test data (see 4.2.1)
(6) Accessory material (see 4.3.2)
(7) Acceptance test reports (see 4.4)
(8) Operational test data (see 4.4.1.2)
(9) Reliability test summary (see 4.4.3.2.1)
(10) Equipment failure (see 4.4.5)
(11) Failure report (see 4.5.4.1)
(12) Test procedures (see 4.6)
(13) Rejection and retest data (see 4.9)
(14) Selection of applicable levels of packaging and packing (see 5.1)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-T-81139(AS) TEST SET, GUIDED MISSILE AN/DSM-68

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Naval Air Systems Command
Washington, D.C. 20360

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